Arithmetic Quantum Cat Maps:
Exact Moment Reductions, Finite Certificates, and the Remaining BGS Obstruction

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Abstract

We give a conservative reconstruction of a long computational programme on arithmetic quantum cat maps. The main verified contribution is an exact pair-key reduction for the fourth-moment geometry. It isolates a weighted diagonal key, decomposes the resulting difference distribution into off/off, mixed, and diagonal pieces, and converts the mixed term through a Cayley transform into an additive-energy problem. An exhaustive integer matching audit and exact cyclotomic resultants give the exceptional prime sets

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\},$$

while the exact design $(p, K) = (461, 23)$ has collision energy 1190 against a forced baseline 902, ruling out a naive uniform-in-K extrapolation. The remaining off/off term admits a promising elliptic-family organization, but the required sheaf, isogeny, bad-fiber, and boundary hypotheses are not yet closed; the associated square-root correlation estimate is therefore a theorem draft, not a theorem. We also record the scope of earlier finite Lean certificates, twisted-Jacobi regressions, antiunitary symmetry calculations, and finite GOE comparisons. In particular, no result in this paper proves microscopic Bohigas–Giannoni–Schmit (BGS) eigenphase statistics.

1 Purpose, conventions, and evidence levels

The project began as an attempt to connect arithmetic structure in quantized hyperbolic toral automorphisms with random-matrix spectral statistics. The historical workspace accumulated correct finite identities, useful numerical experiments, incomplete proof routes, and several false "proved" labels. This manuscript states only the part that survives a reconstruction audit.

Throughout, p denotes an odd prime and $H \leq \mathbf{F}_p^\times$ a cyclic subgroup of order K. Division denotes multiplication by the inverse in $\mathbf{F}_p$. When a statement concerns the audited designs $K = 11, 13, 23$, this is stated explicitly. We use four evidence levels:

- **Exact**: a current algebraic derivation.
- **Finite certificate**: an exhaustive or machine-checked statement at explicitly specified finite parameters.
- **Numerical**: floating-point evidence only.
- **Draft**: a proof architecture with named unresolved obligations.

A finite certificate is not silently upgraded to a uniform theorem, and a numerical fit is not a hypothesis test unless its statistical assumptions are verified.

The relevant established cat-map literature is mainly about eigenfunctions, Hecke symmetries, quantum ergodicity, and matrix-element fluctuations. Kurlberg–Rudnick compute second and fourth moments for normalized matrix elements and compare them with a conjectural SU(2)-trace law [2]; short-window results concern sums of matrix coefficients [3]. Recent higher-dimensional quantum-ergodicity results [5, 6] do not provide a microscopic eigenphase local law. This distinction is load-bearing for the interpretation below.

2 The exact pair-key geometry

Let $x, y \in \mathbf{F}_p \setminus \{0, 1\}$ and set

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y. \tag{1}$$

Proposition 2.1 (Off-diagonal pair-key bijection)**.** *If $x \neq y$, then $u, v \in \mathbf{F}_p^\times \setminus \{1\}$, $u \neq v$, and*

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v}, \qquad d = \frac{(u-1)(v-1)}{v-u}. \tag{2}$$

Conversely every pair $u, v \in \mathbf{F}_p^\times \setminus \{1\}$ with $u \neq v$ arises from exactly one off-diagonal pair (x, y).

Proof. From $x = uy$ and $x - 1 = v(y-1)$ we obtain $(u-v)y = 1 - v$, which gives the first two formulas in (2). Then

$$x - y = (u-1)y = \frac{(u-1)(1-v)}{u-v} = \frac{(u-1)(v-1)}{v-u}.$$

The domain restrictions are immediate from (1); the displayed inverse proves injectivity and surjectivity. □

The diagonal is exceptional in a useful and completely explicit way: all $p - 2$ pairs $x = y \in \mathbf{F}_p \setminus \{0, 1\}$ collapse to the single key

$$(u, v, d) = (1, 1, 0)$$

with multiplicity $p - 2$. Any pair-key calculation that treats this key with multiplicity one is wrong. This weighted diagonal is the mechanism behind several historical factor errors.

3 Fourth-moment coupling and the exact distribution

Let a collapsed key carry its multiplicity $n(u, v)$, equal to one off the diagonal and $p - 2$ at $(1, 1)$. Two ordered keys (u_1, v_1) and (u_2, v_2) are coupled when

$$u_1 u_2 \in H, \qquad v_1 v_2 \in H.$$

Let $B(t)$ be the multiplicity-weighted number of coupled ordered key pairs whose differences sum to t.

Define $O(t)$ to be the off/off contribution and

$$C_H(t) = \#\left\{(u, v) \in (H \setminus \{1\})^2 : u \neq v,\ \frac{(u-1)(v-1)}{v-u} = t\right\}.$$

Then the diagonal bookkeeping is exact:

$$B(t) = O(t) + 2(p-2)C_H(t) + (p-2)^2\mathbf{1}_{t=0}. \tag{3}$$

The factor 2 is forced because diagonal/off and off/diagonal are distinct ordered classes.

Proposition 3.1 (Exact total coupled count)**.** *For the corrected pair-key model,*

$$|V_4| = (K^2+1)p^2 - (6K^2+3K)p + 13K^2 - 3K. \tag{4}$$

Status: Exact formula with finite cross-checks.. The repository verifies (4) by independent implementations. At $(p, K) = (23, 11)$ the total is 48621, while at $(53, 13)$ it is 423879. The Lean certificate at $(53, 13)$ independently checks the diagonal multiplicity, off-diagonal key injectivity, and the latter V_4 count.

4 Character orthogonality and Fourier moments

For the audited character convention, orthogonality collapses the fourth-moment character sum to the finite Fourier transform of B:

$$\sum_{\chi,\psi} |J(\chi,\psi;q)|^4 = m^2 \sum_{t\in\mathbf{F}_p} B(t)e_p(qt), \tag{5}$$

where $e_p(z) = \exp(2\pi iz/p)$ and m is the annihilator dimension in the specified design.

Equation (5) is important because the averaging in q is then deterministic finite Fourier algebra, not a probabilistic independence assumption. If

$$S(q) = \sum_t B(t)e_p(qt),$$

then the mean over $q \neq 0$ follows from

$$\sum_{q\neq 0} e_p(qt) = \begin{cases} p-1, & t = 0, \\ -1, & t \neq 0, \end{cases}$$

and Parseval determines the centered second moment from the L^2 energy of the nonzero-t part of B. Thus the fluctuation problem is reduced to controlling the combinatorial energies in (3).

5 The mixed term as Cayley additive energy

Introduce the Cayley transform

$$c(h) = \frac{h+1}{h-1}, \qquad h \in H \setminus \{1\}.$$

A direct calculation gives

$$c(u) - c(v) = \frac{2(v-u)}{(u-1)(v-1)},$$

hence

$$d(u,v) = \frac{2}{c(u)-c(v)}. \tag{6}$$

Let $A = c(H \setminus \{1\})$. Since $x \mapsto 2/x$ permutes $\mathbf{F}_p^\times$, (6) identifies the nonzero multiplicities of C_H with the nonzero difference multiplicities of A. Therefore

$$\sum_t C_H(t)^2 = E_+(A) - |A|^2, \tag{7}$$

where $E_+(A) = \sum_x r_{A-A}(x)^2$.

Moreover $c(h^{-1}) = -c(h)$, so $A = -A$. This central symmetry forces a baseline of repeated differences. With $|A| = K - 1$ and $0 \notin A$, the collision baseline used by the finite verifier is

$$(K-1)(2K-5). \tag{8}$$

The same differences can be written as intersections of $H \setminus \{1\}$ with a fractional-linear image, giving an exact PGL_2 interpretation of the mixed term. This is useful structurally, but the old attempt to apply a uniform square-root bound directly to the historical Möbius fibers was false: the $d = 0$ fiber degenerates to a constant map.

6 Exact matching audit and cyclotomic exceptional primes

The collision equation between two Cayley differences can be cleared of denominators and expressed in powers of a primitive K-th root. Matching the two six-term multisets gives $6! = 720$ integer linear systems in the exponent variables.

Proposition 6.1 (Finite integer matching classification)**.** *The active exact verifier classifies the 720 systems as*

$$30 \text{ of rank } 2, \qquad 240 \text{ of rank } 3, \qquad 450 \text{ of rank } 4.$$

The gcds of maximal nonzero minors show that the only rational prime at which these integer matrices can acquire an additional rank drop is 2*. Among the rank-*2 *and rank-*3 *kernels, the admissible positive-dimensional systems are exhausted by the identity and inversion-swap symmetries; the others force a forbidden zero or diagonal configuration.*

Status: Exact exhaustive finite algebra.. This is recomputed in CI from the integer matrices rather than accepted from a stored JSON file.

For a specified odd prime K, a normalized non-forced collision gives a cyclotomic polynomial value in $\mathbb{Z}[\zeta_K]$. When this algebraic integer is nonzero, a finite-field collision can occur only at rational primes dividing its norm. The current repository performs the complete normalized nonvanishing/resultant enumeration for $K = 11$ and $K = 13$.

Theorem 6.2 (Audited exceptional sets)**.** *The exact resultant computation gives*

$$E_{11} = \{23, 67, 89\}, \qquad E_{13} = \{53, 79, 131, 157, 313\}.$$

For $K = 11$ *the enumeration checks* 793 *normalized non-forced quadruples and finds* 8 *distinct nonzero norms; for* $K = 13$ *it checks* 1431 *quadruples and finds* 14 *distinct nonzero norms.*

The finite-field regression tests both directions of the prediction:

(p, K)	exceptional?	$\sum_t C_H(t)^2 -$ baseline	result
$(53, 13)$	yes	208	excess observed
$(67, 11)$	yes	48	excess observed
$(199, 11)$	no	0	forced only
$(521, 13)$	no	0	forced only.

The historical verifier had a weaker one-way condition in which an expected exceptional prime passed without requiring observed excess; the active verifier closes that loophole.

6.1 A non-uniformity witness

At $(p, K) = (461, 23)$ the exact finite collision energy is

$$\sum_t C_H(t)^2 = 1190,$$

whereas (8) equals $22(46 - 5) = 902$. The excess is 288. This does not contradict the fixed-parameter cyclotomic mechanism; it shows that a statement calibrated at $K = 11, 13$ cannot be silently promoted to a uniform-in-K bound.

7 The off/off elliptic route

After the exact mixed term is separated, the unresolved part of the moment problem is the centered energy of $O(t)$ in (3). Historical symbolic computations organize the relevant fibers as a one-parameter genus-one family and produce a nonconstant rational j-map, visible degenerate fibers, and same-j parameter symmetries.

Status: Theorem draft, not a theorem.. Nonconstant j proves only non-isotriviality of the candidate generic family. It does *not* by itself imply the trace-function correlation bound needed here. A publication-grade argument must still establish:

1. a smooth projective model over an explicitly specified open base;
2. the complete discriminant and bad-fiber locus, including characteristic exceptions;
3. the relation between the affine counting problem and the deleted projective boundary;
4. purity, lissity, geometric irreducibility, and a uniform conductor/ramification bound for the actual R^1 sheaf;
5. all geometric isomorphisms and twists between parameter fibers;
6. the relevant separable and inseparable isogeny exclusions;
7. the exact hypotheses of the chosen sheaf-correlation theorem.

Michel-type family arguments [7], quantitative sheaf theory [8], Tate/isogeny technology such as [9], and the recent joint-family example of Grove–Saad [10] are relevant tools and templates. They are not plug-in proofs for this pencil.

If the required square-root correlation step is established with uniform constants, the existing combinatorics lead to the target

$$O\Big(K^2 p + K^4 p^{1/2} + K^4\Big). \tag{9}$$

The error in (9) is lower order on the intended main scale in the regime $K = o(p^{1/4})$. Both (9) and this regime remain conditional until the listed geometric obligations are closed.

8 Earlier results retained after the audit

A central aim of the reconstruction was to avoid the opposite errors of trusting every historical "PROVEN" label and deleting every result not on the newest proof route.

8.1 Finite torus geometry at p=199

The active Lean module `TorusLevelSets.lean` verifies for

$$Q(a,b) = a^2 - ab - b^2 \pmod{199}$$

and the cyclic action of $A^T = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$: the action generator has order 11; every nonzero level has 198 nonzero points; the zero cone minus the origin has 396 points; every nonzero level has 18 cyclic orbits; and the action is free on those nonzero levels. These are finite `native_decide` certificates. The analytic Weil intertwining statement that motivated them is not formalized by this file.

8.2 Finite branch scale at p=199

The active Lean module `TorusScaleLaw.lean` checks that $a = 139$ has order 11, the recorded inverse is 63, the half-index trace is 123, and the branch scales 123 and $76 = -123$ satisfy

$$123^2 \equiv 76^2 \equiv 5, \qquad 123 \cdot 76 \equiv -5 \pmod{199}.$$

Again, this is a finite design certificate, not a general scale theorem.

8.3 Twisted-Jacobi finite regressions

The preserved scalar/vectorized checks and q-curve moment records verify the implementation at the stated designs. A historical domain omission was identified: the old closed form requires the relevant $n_1 \neq 0$; on the audited nonzero-q curves this is automatic, but the unrestricted old statement is not retained.

8.4 Antiunitary symmetry: the safe statement

If an antiunitary Θ obeys $\Theta U \Theta^{-1} = U^{-1}$ and $Uv = \lambda v$ with $|\lambda| = 1$, then anti-linearity gives

$$U\Theta v = \bar{\lambda}^{-1}\Theta v = \lambda \Theta v.$$

Thus the relevant eigenspace is preserved. An earlier objection that Θ necessarily sends it to a distinct λ^{-1} sector forgot the conjugation from anti-linearity.

What is *not* promoted is a blanket claim that every relevant global block has already been certified as GOE class. The historical finite/kernel chart remains useful; a full blockwise statement requires a fresh treatment of all charts and conventions if it becomes load-bearing.

8.5 Shape, Catalan, and arcsine material

The archive contains exact finite relation counts and moment patterns that motivated Catalan/arcsine conjectures. They remain useful for discovery and regression. We do not promote the historical all-orders full-group arcsine package because its word-trace reduction has an unresolved higher rewrite case. Agreement of finitely many moments is not an all-orders proof.

9 Finite spectral diagnostics

Finite parity-reduced spectra at the recorded designs show descriptive agreement with random-matrix benchmarks. The current repository deliberately downgrades two historical statistical labels.

First, the spacing record now reports only the empirical CDF sup-distance to the GOE Wigner surmise. The historical script compared this number with an IID Kolmogorov–Smirnov threshold, but gaps obtained after polynomial unfolding and pooled within the same spectra are dependent. We therefore do not report a KS "pass".

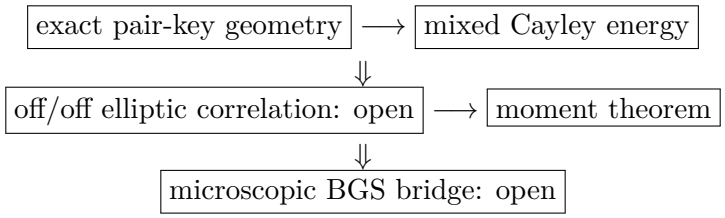
Second, the old file called "number variance" used forward windows starting at each eigenvalue and truncated at the spectral edge. This is not the standard stationary number variance $\Sigma^2(L)$. The values are retained as a matched-GOE window-count diagnostic because the cat-map and GOE samples were processed by the same algorithm, but the old two-sigma pass/fail interpretation is withdrawn.

These numerical observations are compatible with the random-matrix motivation and with earlier numerical cat-map studies [4]; they are not evidence for a limiting local law.

10 Why the BGS problem remains open

The BGS claim concerns microscopic eigenphase correlations after all relevant symmetries are removed. The exact pair-key reduction and even the prospective estimate (9) are moment statements about a structured arithmetic family. A local-statistics theorem needs a separate mechanism that controls eigenvalue correlations at the mean-spacing scale. No such transfer theorem has been verified for the present family.

Accordingly, the current logical chain is



Closing an earlier box does not silently close a later one.

11 Reproducibility and negative controls

The active repository keeps a small scientific surface:

- Python exact regressions and finite-field primitives;
- two active C++ programs: an independent fourth-moment pair-key verifier and an exact cat-map order/split-prime utility;
- six active Lean finite-certificate modules;
- promoted finite certificates and explicitly descriptive numerical data;
- a separate historical record for retractions and old experiments.

The $K = 11, 13$ Cayley certificate is regenerated in CI and compared structurally with the committed JSON. The C++ verifier independently checks the pair-key geometry at finite designs. CI also checks that every active Lean module is imported before running `lake build`.

Negative controls are retained when they explain a failure mode. In particular, the degenerate historical Möbius fiber is formalized as a finite falsifier, and $(461, 23)$ is retained as a non-uniformity witness. Historical Julia source is inventoried for later migration; false algorithms are not translated merely to preserve a language port.

12 Retractions that affect interpretation

The following historical conclusions are explicitly not part of the present result:

- the Möbius/M4D proof of the mixed/off-diagonal fluctuation bound;
- the one-sided mixed-class count that omitted one of the two ordered diagonal/off classes;
- any shortcut from a global random-matrix universality theorem to this structured deterministic arithmetic family without verified hypotheses;
- the historical unrestricted twisted-Jacobi formula that omitted its domain condition;
- the all-orders arcsine/Catalan conclusion based on the unresolved word-trace rewrite;
- finite numerical GOE "passes" interpreted as proof of a limiting law.

These failures are kept in the research record because they are useful regression tests; they are excluded from the active theorem surface.

13 Conclusion

The reconstruction leaves a substantially narrower but stronger result than the historical project claimed. The pair-key geometry, weighted diagonal, fourth-moment coupling, Fourier reduction, Cayley transform, integer matching audit, the explicit E_{11} and E_{13} resultant sets, and the $(461, 23)$ obstruction are reproducible. Earlier finite torus, scale, and twisted-Jacobi results survive with their actual finite scope.

The main analytic task is now sharply localized: turn the off/off elliptic organization into a theorem by verifying the precise sheaf and isogeny hypotheses. Even after that, microscopic BGS statistics require a separate local-law bridge. Keeping these closure conditions explicit is more useful than a stronger but unsupported theorem statement.

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